Precursor (Example: phenyl acetic acid) is added to the medium

↓

Antifoam agent (Example: corn or soybean oil) is added before sterilization

The sufficient aeration and agitation is given and are incubated at 25°C to 26°C for 3 to 5 days at PH range of 7 to 7.5

Penicillin Production

Process of penicillin production occurs in three phases:

First phase: Growth of mycelium occurs in this phase where the yield of antibiotic is low. The pH increases due to the release of NH_3.

Second phase: In this phase, intense synthesis of penicillin occurs due to rapid consumption of Lactose and Ammonium nitrogen. The mycelial mass increases and the pH remain unchanged (Figure 6.10).

Third phase: In this phase, the concentration of antibiotics decreases in the medium. Autolysis of mycelium starts, liberating Ammonia leading to slight rise in pH.

Recovery

After penicillin fermentation, the broth is filtered on rotary vacuum filter

↓

Mycelium is separated

↓

To the broth sulphuric acid or phosphoric acid is added

↓

Penicillin is converted into anionic form

↓

It is extracted in counter current solvent extractor, by using organic solvent, amyl acetate, methyl isobutyl (ketone)

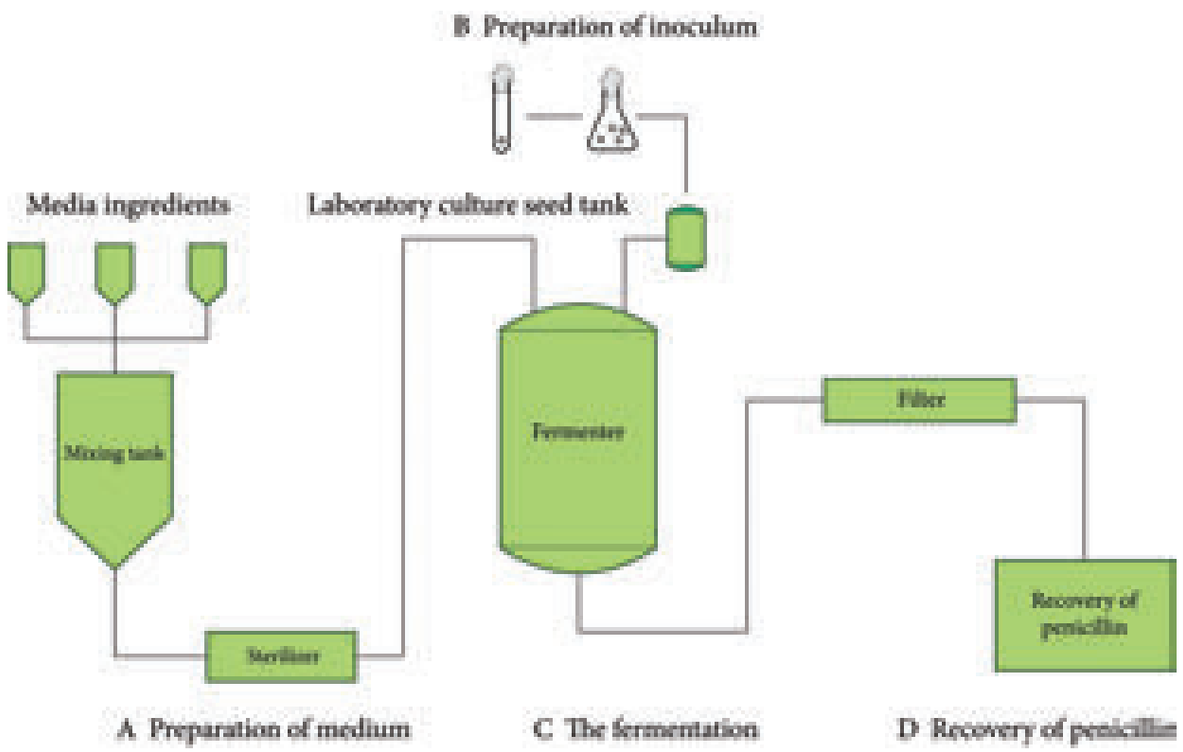


Figure 6.10: Production of Penicillin

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